COMPARISON OF BOTTLED WATER PRICES BY BOTTLE SIZE AND BRAND IN MALAYSIA

QMQ27103

ENGINEERING STATISTICS

ASSIGNMENT REPORT

UNIVERSITI MALAYSIA PERLIS

Dataset: 150 approximate product–store records

How do bottled water prices vary between different brands and bottle sizes in Malaysia, and which bottle size and brand provides the most affordable option?

Investigation Question:

Submission Coverage: Week 1–Week 3

Semester: 1

Academic Session: 2026/2027

Course Code: QMQ27103 — Engineering Statistics

Student ID: 241090104-5

Student Name: Redhwan Ali Hasan Abdullah

WEEK 1 — From Problem to Data

1. Your Interest and Initial Ideas

I am interested in comparing bottled-water prices in Malaysia because bottled water is commonly used by students and the public, and many brands and bottle sizes are available. I want to understand how prices change across brands and sizes and identify more affordable options.

2. Look Closer: What Could Vary?

- Prices may differ by brand (for example, Spritzer, Bleu, Evian and others).
- Prices may differ by bottle size, from small bottles to large containers.
- Prices may vary between online store labels.
- Water type (such as mineral, drinking or distilled) may be related to price.
- Larger bottles may offer a lower price per litre than smaller bottles.

3. Focus and Investigation Question

The investigation focuses on comparing price per litre across brands, sizes and store labels, and identifying the highest- and lowest-priced records in the prepared dataset.

Main question: How do bottled water prices vary between different brands and bottle sizes in Malaysia, and which bottle size and brand provides the most affordable option?

4. Problem and Purpose

Consumers may find it difficult to compare products using package prices alone because bottles have different volumes. Converting price to RM per litre creates a common unit for comparison. The analysis is intended to describe variation and support more informed comparisons, rather than claim a universal cheapest brand.

5. Data Plan

- Population of interest: bottled-water products offered to consumers in Malaysia.
- Observational unit: one product–store record (a brand/size candidate paired with a store label).
- Dataset size: 150 records.
- Qualitative variables: Brand, Online store, Water type, Product label.
- Quantitative variables: Bottle size (mL), Price (RM), Price per litre (RM/L).
- Planned measure: Price per litre = Price (RM) ÷ Bottle size (mL) × 1,000.

6. Dataset Generation and Important Limitation

The working dataset contains 30 brand/size candidates repeated across five store labels, yielding 150 rows. The values are approximate estimates prepared for practice. Some product–store combinations may not be available, and the URLs/date-checked fields are not verified listing evidence. This limits representativeness and means the results must not be presented as confirmed current store prices.

7. Objectives

1. Describe the number of product records listed for each brand.
2. Explore the distribution of price per litre.
3. Compare the five highest and five lowest price-per-litre records.
4. Compare price-per-litre distributions across the five store labels.
5. Use observed variation and uncertainty to state cautious conclusions.

WEEK 2 — From Data to Evidence

1. Prepare and Describe the Data

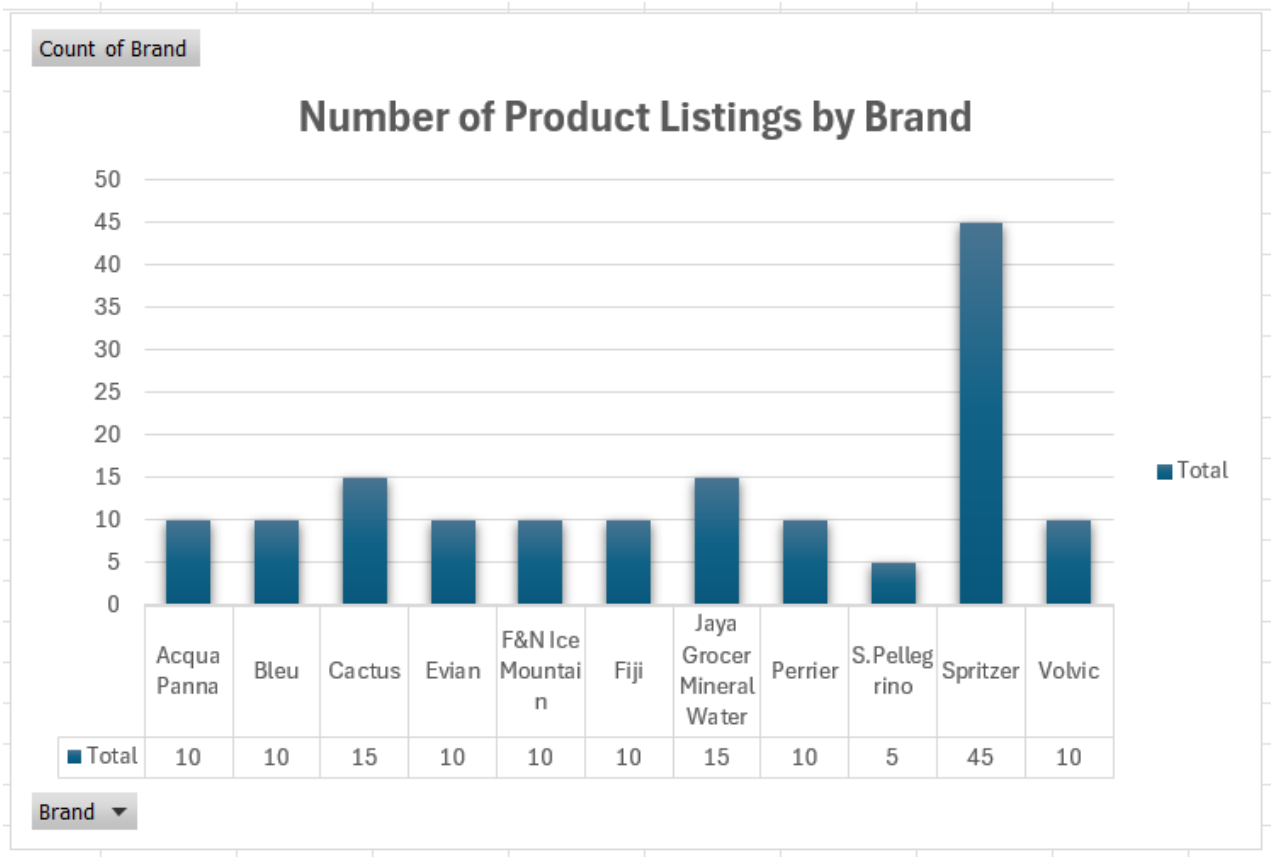
The dataset contains 150 product–store records. Each row represents a candidate brand and bottle size associated with one of five store labels: Jaya Grocer, Servay Online, Lotus's, AEON and NSK Grocer. Variables include brand, bottle size, price, store, price per litre, product label and water type.

Price per litre is used to make bottle sizes comparable. Package price and unit price answer different questions: the former is the amount paid for one bottle, while the latter supports comparison of value by volume.

2. Graphical Summaries

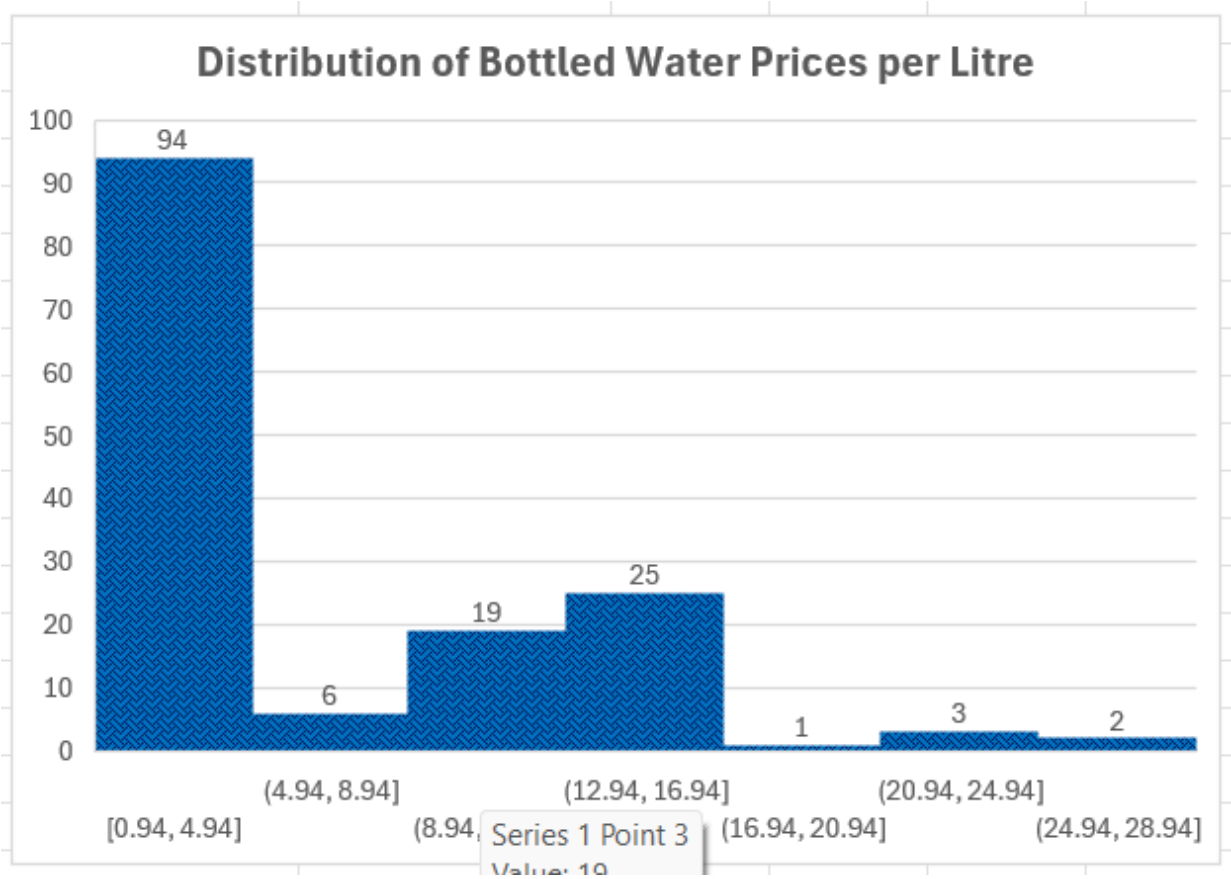
2.1 Qualitative variable — Brand listing frequency

A horizontal/brand frequency chart summarizes the number of product records associated with each brand. The displayed counts total 150.



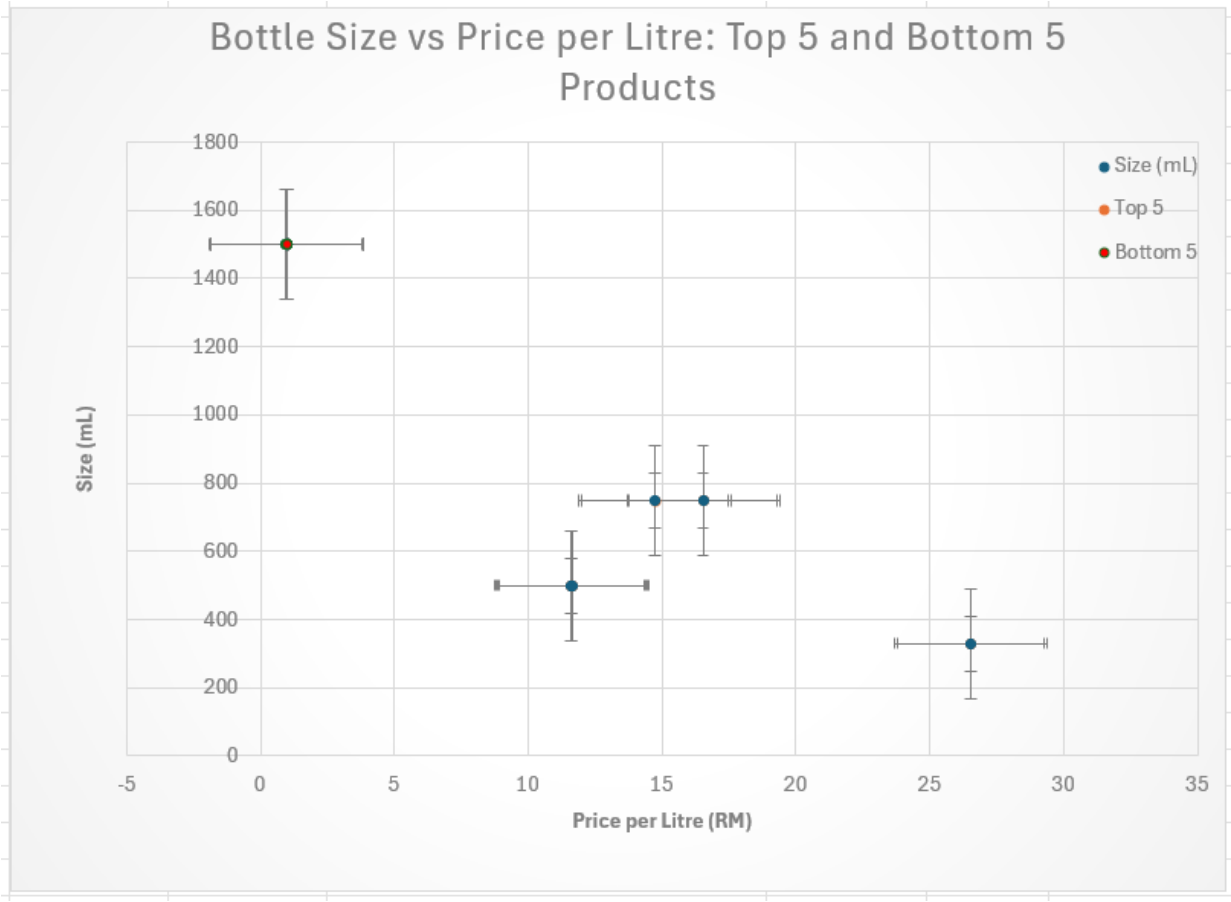
Interpretation: Spritzer has the highest frequency (45 records, 30% of the dataset). Cactus and Jaya Grocer Mineral Water each have 15 (10%). Most other brands have 10 each (6.7%), while S.Pellegrino has 5 (3.3%). These are dataset record counts, not market-share estimates.

2.2 Quantitative variable — Histogram of price per litre



The histogram contains 150 observations. The first bin, RM0.94–RM4.94/L, contains 94 records; the next bins contain 6, 19, 25, 1, 3 and 2 records. Most observations lie in the first interval, and the remaining observations form a long upper-price tail. The distribution is right-skewed, with relatively few expensive observations.

2.3 Top-five and bottom-five comparison



The selected high-price records range from RM11.61/L to RM26.54/L, while the selected low-price records are around RM0.94–RM1.006/L. The shown values indicate a large difference in unit price. This comparison is descriptive and depends on how products were selected and on the approximate input values.

Chart quality note: the supplied scatter chart appears to have an unclear series setup and error bars, so it does not communicate the intended ten product points cleanly. Treat the numeric comparison as the main evidence; revise the chart to plot bottle size (mL) on X and price per litre (RM/L) on Y, with only the ten selected records and product labels.

2.4 Box-and-whisker plot — Price per litre by store label



The five store distributions overlap considerably, and each shows a broad range. The plot suggests substantial variation within each store label. The current Excel chart groups all five boxes under one x-axis category ("1") and uses the legend for store names; this is readable but could be improved by displaying store names directly beneath the boxes. Because prices were estimated and the same candidate list was replicated across store labels, this plot cannot establish actual retailer price differences.

3. Numerical and Qualitative Summaries

Qualitative summary: the dataset is unbalanced by brand. Spritzer accounts for 45 records; S.Pellegrino accounts for 5.

Distribution summary: the histogram shows concentration in RM0.94–RM4.94/L and a sparse high-price tail. The displayed extreme selected values include RM0.94/L at the lower end and RM26.54/L at the upper end.

The prepared dataset is designed with repeated brand/size candidates across stores and approximate store adjustments. This structure should be considered when interpreting apparent differences and variability. Inferential claims and market-wide conclusions are not justified.

4. Interpret the Evidence

The charts indicate that unit prices differ greatly across product records. The high concentration in the first histogram interval suggests many records are comparatively inexpensive per litre, while a small number of premium-price records extend the range. Brand record counts are uneven, but frequency only describes the dataset design and does not indicate popularity or availability.

The analysis supports using RM/L instead of bottle price alone when comparing value. It does not establish that a particular brand or bottle size is always cheapest because the values are estimates, water types and product specifications may differ, and the candidate list is limited.

5. Connection Across Graphs

The bar chart describes which brands contribute the most records; the histogram describes the overall distribution of unit prices; the selected-record comparison highlights extremes; and the box plot explores spread across store labels. Together, they show that variability has several dimensions. However, their conclusions are constrained by the same approximate dataset.

WEEK 3 — Reason with Data

A. Observe: Understanding Variability

1. Chart 1 — Histogram

The histogram reveals strong right skew: 94 out of 150 records (62.7%) fall between RM0.94 and RM4.94/L. Only a few records occur in the highest intervals, indicating that expensive unit-price observations are uncommon within this constructed dataset.

2. Chart 2 — Box Plot

The box plots show wide price variation within each store group, with similar central regions and overlapping distributions. Their ranges extend far above the lower quartiles, consistent with a right-skewed distribution. Store-to-store distinctions are difficult to interpret because the candidate products were replicated across groups.

3. Compare and Think

The histogram shows the shape and concentration of all unit prices, while the box plot summarizes median, quartiles and spread by store label. Both suggest substantial right-skew and within-group variability. The histogram gives an overall picture; the box plot is intended to compare groups, but the constructed data and chart labeling limit retailer-level interpretation.

B. Explain: Possible Explanations

4. Which variables might explain the variation?

Potential explanatory variables include brand, bottle size, store label and water type. Bottle size matters because unit price may change with volume; brand and water type may relate to positioning or production; store label may be associated with pricing policies. The present dataset does not verify these effects.

5. What other variables might explain variation?

Other useful variables could include promotion/discount status, multipack versus single-bottle price, delivery charges, product origin, packaging, exact product code, date and time of price capture, stock availability, and whether the price is regular or promotional.

C. Reason: Evidence, Uncertainty and Conclusion

6. Evidence — Most important finding

The most visible result is the wide range of unit prices: selected records extend from approximately RM0.94/L to RM26.54/L. In addition, 94 of 150 records lie in the lowest histogram interval. This supports the conclusion that unit price varies considerably across the candidate records.

7. Uncertainty — What remains unknown?

Actual shelf or online prices may differ from the estimates. Some listed brand–size combinations may not be sold by the assigned store. The records do not provide verified URLs, collection dates or promotion details. Repeating the same candidate set across store labels may make groups appear more comparable than real inventories are. The dataset is not a random sample, so uncertainty about the wider Malaysian market cannot be quantified reliably.

8. Conclusion or Decision

For the constructed dataset, comparing price per litre is a more meaningful way to assess affordability than comparing package prices alone. The charts show uneven brand-record counts and a right-skewed unit-price distribution, with a small number of expensive records. No definitive market-wide cheapest brand or retailer can be recommended from these approximate data. A stronger next step is to collect dated, verifiable prices for matching products and volumes across stores, recording promotions and availability.

Key Idea

Variability helps explain why observations differ. Combining graphical evidence with consideration of uncertainty leads to more careful conclusions and better-informed decisions.

D. Develop and Refine Thinking with AI

9. Think First — Initial interpretation

My initial expectation was that larger bottles would usually have a lower price per litre and that premium brands would cost more. I also expected prices to vary across stores. These expectations are ideas to examine, not conclusions.

10. Ask AI — Review and feedback

AI-assisted review helped organize the investigation around comparable units, improve the chart sequence, and identify caveats about estimated prices, repeated candidate records, and unclear chart configuration.

11. Evaluate — Useful feedback

The most useful feedback was to use price per litre for fairer size comparisons and to state that estimated values cannot support claims about actual store prices. The chart review also identified the need to improve scatter-plot series and box-plot category labels.

12. Refine and Reflect — Thinking shift

My thinking shifted from looking for the cheapest bottle price to comparing unit price and examining the distribution. I also learned that a chart can look convincing while its dataset construction or visual encoding limits what it can prove. I would retain the RM/L measure and four-chart structure, but improve the evidence by collecting verified listings and redesigning the scatter plot and box-plot labels.

Appendix: Chart Inventory

Figure 1. Number of product listings by brand. Figure 2. Distribution of bottled-water prices per litre. Figure 3. Top-five and bottom-five selected records (chart needs refinement). Figure 4. Price-per-litre distribution by store label (category labels could be improved).

Data transparency statement: the charts in this report were inserted from the Excel screenshots supplied for this assignment. All interpretations are restricted to the prepared approximate dataset.